

TAJWARUL HASAN CHOWDHURY

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Aspiring Software Engineer with hands-on experience building full-stack and AI-powered applications using TypeScript, JavaScript, React, Next.js, Node.js, PostgreSQL, and MongoDB. Strong foundation in data structures, algorithms, databases, backend development, authentication, and API design. Built end-to-end projects involving AI integration, RAG pipelines with LangChain, asynchronous processing, security, and third-party services, with a focus on solving practical engineering problems and writing maintainable software.

TECHNICAL SKILLS

Languages: C++, JavaScript, TypeScript

Frontend: React, Next.js, HTML5, CSS3, Tailwind CSS, shadcn/ui

Backend: Node.js, Express.js, REST APIs, Zod (schema validation)

Databases & ORM: PostgreSQL, MongoDB, Drizzle ORM, Mongoose, Neon

Authentication & Security: JWT, Better Auth, Firebase Authentication, Role-Based Access Control (RBAC)

AI & LLM: LangChain, Retrieval-Augmented Generation (RAG), Qdrant, LLM API Integration (Groq, Google Gemini)

Deployment & Tools: Git, GitHub, Vercel, Render, Cloudinary

Core CS: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Computer Networks, Database Management Systems (DBMS), System Design

PROJECTS

HireFlow | AI-Assisted Hiring Platform

React, Express 5, MongoDB (Mongoose), Groq, Puppeteer, Firebase Auth, Cloudinary | [GitHub](#) · [Live](#)

- Two-sided hiring platform. Candidates generate AI-written resume PDFs, get a job-specific readiness report (match score, skill gaps, 7-day prep roadmap, tailored technical and behavioral questions), and apply. Companies post jobs with AI-drafted descriptions, invite recruiters, and review applicants ranked by AI match score through an applied → shortlisted → interview → hired pipeline with email and in-app notifications.
- Built the AI features on Groq with strict JSON-schema output re-validated by Zod. Slow work (application scoring, Puppeteer PDF rendering, Cloudinary upload) runs after the API responds, tracked through status fields and client polling. MongoDB indexes support full-text job search, cursor pagination, and a unique (candidate, job) index that blocks duplicate applications.
- Secured the platform with four-role RBAC and company-scoped authorization, JWT in httpOnly cookies with TTL-indexed token blacklisting, Firebase Google sign-in, SHA-256-hashed single-use recruiter invite links, and per-user rate limits on AI, upload, and email endpoints.

Scam Scanner | AI Scam Evidence Investigator

Next.js 16, TypeScript, PostgreSQL (Neon), Drizzle ORM, BullMQ, Redis, Gemini, Groq | [GitHub](#) · [Live](#)

- Lets users upload chat screenshots, PDFs, DOCX files, and pasted messages from a suspected scam and get back a plain-language report: an overall risk level, contradictions between the evidence (quoting each conflicting statement with its source), and 2–4 concrete verification steps, saved to a private report history.
- Engineered the analysis as an async BullMQ pipeline on a standalone Render worker, avoiding serverless time limits. Vision-LLM OCR (multimodal Gemini 2.5 Flash, with Groq fallback) extracts facts into a 12-field typed schema; a rule engine and URL analyzer (RDAP domain age, homograph and brand-impersonation domains, Google Safe Browsing) set a risk floor the LLM cannot lower.
- Hardened handling of untrusted input with magic-byte file checks, DOCX zip-bomb detection, prompt-injection-resistant prompts, Zod-validated model output, Better Auth sessions, and per-user rate limits (Upstash Redis).

SEC Library Management System | Team of 2

React, Express 5, MongoDB (Mongoose), LangChain, Qdrant, Groq, SSLCommerz | [Student Portal](#) · [Admin Portal](#) · [GitHub](#)

- Built the student and admin React portals (catalog search, borrowing, reservations, waitlists, fines, research-paper approval) and Firebase Google sign-in that links a Google account to a pre-registered student record.
- Integrated SSLCommerz payments for fine clearing with server-side transaction validation (never trusting redirect or IPN bodies) and atomic, idempotent updates so duplicate gateway callbacks cannot apply a payment twice.
- Built a RAG catalog chatbot: book data chunked with LangChain and embedded with Gemini into Qdrant; retrieved context goes to a Groq LLM under a strict JSON contract, with per-thread memory so follow-ups resolve to earlier books.

EDUCATION

Sylhet Engineering College | B.Sc. in Computer Science and Engineering | CGPA: 3.69 / 4.00

Feb 2022 – Aug 2026

ACHIEVEMENTS

- Codeforces max rating 1267 (Pupil), handle: [tajwarul](#)
- Solved 1,300+ algorithmic and data structure problems across multiple online programming platforms.
- Participated in 40+ competitive programming contests, strengthening problem-solving and algorithmic skills.